Problem M2: Rapidity in an Asymmetric Collider

Worked Solution with Teaching Commentary

Midterm Preparation Notes

Preamble: Key Definitions and Tools

Before diving in, let's collect the essential machinery.

Rapidity y

For a particle with energy E and longitudinal momentum p_z:

$$y = \frac{1}{2}\ln\frac{E+p_z}{E-p_z} \tag{1}$$

Equivalently, writing $E = m_T \cosh y$ and $p_z = m_T \sinh y$ where $m_T = \sqrt{m^2 + p_T^2}$ is the transverse mass.

Hyperbolic Identities for Beam Particles

For a beam particle moving along $+z$ with rapidity y:

$$\cosh y = \gamma, \qquad \sinh y = \beta\gamma, \qquad \tanh y = \beta \tag{2}$$

The four-momentum can be written as ($p_T = 0$ for beams):

$$p^\mu = (E, 0, 0, p_z) = m(\cosh y,\ 0,\ 0,\ \sinh y) \tag{3}$$

For **beam particles** where $E \gg m$ (i.e. $E \approx |p|$), the rapidity becomes very large:

$$y_{\text{beam}} \approx \frac{1}{2}\ln\frac{2E}{m^2/2E} = \ln\frac{2E}{m} \tag{4}$$

The Crucial Additive Property of Rapidity

Under a Lorentz boost along z with boost rapidity y_{boost}, every particle's rapidity shifts by a constant:

$$\boxed{y' = y - y_{\text{boost}}} \tag{5}$$

This is why rapidity is so useful at colliders — rapidity *differences* are Lorentz invariant.

Proof sketch: Under a boost, $E' = \gamma(E - \beta p_z)$ and $p'_z = \gamma(p_z - \beta E)$. Then:

$$y' = \frac{1}{2}\ln\frac{E'+p'_z}{E'-p'_z} = \frac{1}{2}\ln\frac{(1-\beta)(E+p_z)}{(1+\beta)(E-p_z)} = y + \frac{1}{2}\ln\frac{1-\beta}{1+\beta}$$

The extra term $\frac{1}{2}\ln\frac{1-\beta}{1+\beta} = -y_{\text{boost}}$ (by the definition of rapidity applied to the boost itself), giving $y' = y - y_{\text{boost}}$.

Pseudorapidity η

$$\eta = -\ln \tan \frac{\theta}{2} \tag{6}$$

where θ is the polar angle from the beam axis. For massless or highly relativistic particles $(m \ll p)$, $\eta \approx y$.

1 Part (a): y_{CM} from the Total Four-Momentum

What we need to show

The rapidity of the center-of-mass frame y_{CM} equals the rapidity computed from the total four-momentum $P^\mu = p_a^\mu + p_b^\mu$.

Setup

Two beam particles, both protons (mass m), move along the z-axis. Using the hyperbolic parameterization (3):

$$p_a^\mu = m(\cosh y_a,\ 0,\ 0,\ +\sinh y_a) \qquad \text{(beam } a\text{, moving in } +z) \tag{7}$$

$$p_b^\mu = m(\cosh y_b,\ 0,\ 0,\ +\sinh y_b) \qquad \text{(beam } b\text{, moving in } -z) \tag{8}$$

Note: beam b moves in $-z$, so $\sinh y_b < 0$ (equivalently, $y_b < 0$).

Total four-momentum

$$P^\mu = p_a^\mu + p_b^\mu = m\big(\cosh y_a + \cosh y_b,\ 0,\ 0,\ \sinh y_a + \sinh y_b\big) \tag{9}$$

Rapidity of P^μ

Apply definition (1) with $E_{\text{tot}} = m(\cosh y_a + \cosh y_b)$ and $p_{z,\text{tot}} = m(\sinh y_a + \sinh y_b)$:

$$y_{\text{CM}} = \frac{1}{2}\ln\frac{E_{\text{tot}} + p_{z,\text{tot}}}{E_{\text{tot}} - p_{z,\text{tot}}} = \frac{1}{2}\ln\frac{(\cosh y_a + \cosh y_b) + (\sinh y_a + \sinh y_b)}{(\cosh y_a + \cosh y_b) - (\sinh y_a + \sinh y_b)} \tag{10}$$

Use $\cosh y + \sinh y = e^y$ and $\cosh y - \sinh y = e^{-y}$:

$$y_{\text{CM}} = \frac{1}{2}\ln\frac{e^{y_a} + e^{y_b}}{e^{-y_a} + e^{-y_b}} \tag{11}$$

Why this is y_{CM}

The CM frame is defined as the frame where $p'_{z,\text{tot}} = 0$. Since rapidity is additive under boosts (5), we need to boost by exactly the amount that zeros out the total p_z. The rapidity of the total four-momentum is precisely that boost — it's the rapidity of the "system as a whole."

Part (a) Result

$$\boxed{y_{\text{CM}} = \frac{1}{2}\ln\frac{e^{y_a} + e^{y_b}}{e^{-y_a} + e^{-y_b}}} \tag{12}$$

2 Part (b): Equal Masses $\Rightarrow y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$

Simplification

Since $m_a = m_b$ (both protons), the masses already cancelled in (12). Now factor:

$$y_{\text{CM}} = \frac{1}{2} \ln \frac{e^{y_a} + e^{y_b}}{e^{-y_a} + e^{-y_b}} \tag{13}$$

Multiply numerator and denominator by $e^{(-y_a - y_b)/2}$... actually, let's do it more cleanly. Factor $e^{y_a/2 + y_b/2}$ from the numerator and $e^{-y_a/2 - y_b/2}$ from the denominator:

Numerator:

$$e^{y_a} + e^{y_b} = e^{(y_a+y_b)/2} \left(e^{(y_a-y_b)/2} + e^{-(y_a-y_b)/2} \right) = e^{(y_a+y_b)/2} \cdot 2 \cosh \frac{y_a - y_b}{2}$$

Denominator:

$$e^{-y_a} + e^{-y_b} = e^{-(y_a+y_b)/2} \left(e^{-(y_a-y_b)/2} + e^{(y_a-y_b)/2} \right) = e^{-(y_a+y_b)/2} \cdot 2 \cosh \frac{y_a - y_b}{2}$$

The cosh factors cancel:

$$y_{\text{CM}} = \frac{1}{2} \ln \frac{e^{(y_a+y_b)/2}}{e^{-(y_a+y_b)/2}} = \frac{1}{2} \ln e^{y_a + y_b} \tag{14}$$

Part (b) Result

$$\boxed{y_{\text{CM}} = \frac{1}{2}(y_a + y_b)} \tag{15}$$

The CM rapidity is the **average** of the two beam rapidities. For a symmetric collider ($E_a = E_b$), $y_a = -y_b$ and $y_{\text{CM}} = 0$, i.e. the lab *is* the CM frame.

3 Part (c): Beam Rapidities in the CM Frame

The boost

To go from the lab to the CM frame, we boost by $y_{\text{boost}} = y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$. Using the additive property (5):

Beam a in the CM frame:

$$y_a^* = y_a - y_{\text{CM}} = y_a - \frac{1}{2}(y_a + y_b) = +\frac{1}{2}(y_a - y_b) \tag{16}$$

Beam b in the CM frame:

$$y_b^* = y_b - y_{\text{CM}} = y_b - \frac{1}{2}(y_a + y_b) = -\frac{1}{2}(y_a - y_b) \tag{17}$$

Part (c) Result

$$\boxed{y_a^* = +\frac{1}{2}(y_a - y_b), \qquad y_b^* = -\frac{1}{2}(y_a - y_b)} \tag{18}$$

The beams have **equal and opposite** rapidities in the CM frame, as expected by symmetry. The CM frame is symmetric even when the lab frame isn't.

Physical check

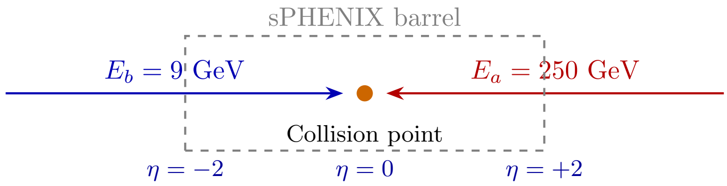
For a symmetric collider (say LHC, $E_a = E_b = 7$ TeV): $y_a \approx +8.8$, $y_b \approx -8.8$. Then $y_a^* = +8.8$, $y_b^* = -8.8$ — same as the lab, because the lab *is* the CM frame. Good.
For a fixed-target experiment ($E_b = m$, $y_b = 0$): $y_a^* = y_a/2$, $y_b^* = -y_a/2$. The available rapidity range is halved compared to the lab — energy is "wasted" on the CM motion.

4 Part (d): The RHIC Asymmetric Run and sPHENIX

The physical situation

RHIC runs asymmetric proton beams to probe forward rapidities:

- Beam a: $E_a = 250$ GeV (moving in $+z$)
- Beam b: $E_b = 9$ GeV (moving in $-z$, the minimum "booster" energy)
- Proton mass: $m_p = 0.9383$ GeV



Step 1: Compute $\sqrt{s}$

For colliding beams, using $E \gg m$ so $|\vec{p}| \approx E$:

$$s = (p_a + p_b)^2 = m_a^2 + m_b^2 + 2(E_a E_b + |\vec{p}_a||\vec{p}_b|) \tag{19}$$

Since both beams move along z in *opposite* directions and $|\vec{p}| \approx E$:

$$s \approx 2E_a E_b + 2E_a E_b = 4E_a E_b \tag{20}$$

Wait — let's be more careful. With $p_a^\mu \approx (E_a, 0, 0, +E_a)$ and $p_b^\mu \approx (E_b, 0, 0, -E_b)$:

$$s = (E_a + E_b)^2 - (E_a - E_b)^2 = 4E_a E_b \tag{21}$$

$\sqrt{s}$

$$\boxed{\sqrt{s} = 2\sqrt{E_a E_b} = 2\sqrt{250 \times 9} = 2\sqrt{2250} \approx 94.9 \text{ GeV}} \tag{22}$$

Step 2: Compute y_a and y_b

For beam particles with $E \gg m$, moving along $+z$:

$$y_a = \frac{1}{2} \ln \frac{E_a + p_a}{E_a - p_a} \tag{23}$$

With $p_a = \sqrt{E_a^2 - m^2} \approx E_a(1 - m^2/2E_a^2)$:

$$y_a \approx \frac{1}{2} \ln \frac{2E_a}{m^2/2E_a} = \ln \frac{2E_a}{m} = \ln \frac{500}{0.9383} = \ln(532.9) \tag{24}$$

$$\boxed{y_a \approx 6.28} \tag{25}$$

For beam b moving in $-z$, $p_z = -|\vec{p}_b| \approx -E_b$:

$$y_b = \frac{1}{2} \ln \frac{E_b - E_b}{E_b + E_b} \approx -\ln \frac{2E_b}{m} = -\ln \frac{18}{0.9383} = -\ln(19.18) \tag{26}$$

$$\boxed{y_b \approx -2.95} \tag{27}$$

Sanity check

For a symmetric 250+250 collider, $y_a = +6.28$ and $y_b = -6.28$, giving $y_{\text{CM}} = 0$. Here the 9 GeV beam has a much smaller rapidity magnitude, so the CM is shifted toward it.

Step 3: Compute y_{CM}

Using (15):

$$y_{\text{CM}} = \frac{1}{2}(y_a + y_b) = \frac{1}{2}(6.28 - 2.95) = \frac{1}{2}(3.33) \tag{28}$$

y_{CM}

$$\boxed{y_{\text{CM}} \approx +1.66} \tag{29}$$

The CM frame is boosted by $y = 1.66$ in the $+z$ direction (toward the high-energy beam). This makes sense — the 250 GeV beam carries far more momentum, so the CM frame "chases" it.

Cross-check with $\sqrt{s}$: We can also compute y_{CM} from:

$$y_{\text{CM}} = \frac{1}{2}\ln\frac{E_a}{E_b} = \frac{1}{2}\ln\frac{250}{9} = \frac{1}{2}\ln(27.78) = \frac{1}{2}(3.325) = 1.66 \quad \checkmark$$

(This shortcut works because for $E \gg m$ beam particles, $y_a \approx \ln(2E_a/m)$ and $y_b \approx -\ln(2E_b/m)$, so $y_a + y_b \approx \ln(E_a/E_b)$.)

Step 4: Translate the sPHENIX Acceptance

Symmetric case (reference)

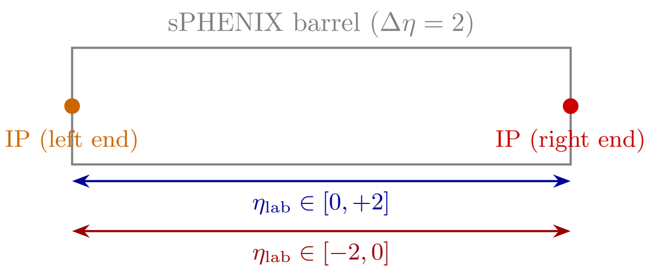
In the normal symmetric mode, collisions happen at the **center** of the barrel, and the sPHENIX Central Arm covers:

$$-1 < \eta < +1 \qquad \text{(lab frame, symmetric collisions)}$$

Since the lab = CM frame in symmetric mode, this is also $-1 < y_{\text{CM}} < +1$.

Asymmetric case: collision at the ends of the barrel

The problem states that the collision point is moved to either **end** of the barrel. The barrel itself has a total $\Delta\eta$ span of 2 (this is a fixed geometric property). The detector acceptance in the lab does *not* depend on beam energies, only on where the collision occurs relative to the barrel.



- **Collision at the left end of barrel:** the detector extends to the *right* of the IP, covering $\eta_{\text{lab}} \in [0, +2]$.
- **Collision at the right end of barrel:** the detector extends to the *left* of the IP, covering $\eta_{\text{lab}} \in [-2, 0]$.

Translating to CM-frame rapidity

The problem tells us that for the detected particles, $\eta \approx y$ (they are relativistic enough that $m \ll p$). To convert lab-frame η to CM-frame y, we simply subtract y_{CM}:

$$y^* = y_{\text{lab}} - y_{\text{CM}} = \eta_{\text{lab}} - y_{\text{CM}} \tag{30}$$

Translated Acceptance Limits

Collision at left end ($\eta_{\text{lab}} \in [0, +2]$):

$$\boxed{y^* \in [0 - 1.66,\ +2 - 1.66] = [-1.66,\ +0.34]} \tag{31}$$

Collision at right end ($\eta_{\text{lab}} \in [-2, 0]$):

$$\boxed{y^* \in [-2 - 1.66,\ 0 - 1.66] = [-3.66,\ -1.66]} \tag{32}$$

Physical interpretation — why this is the whole point

In the symmetric 250+250 collider, sPHENIX covers $y^* \in [-1, +1]$, i.e. **mid-rapidity**. By going to asymmetric beams and moving the collision point, we now access:

- $y^* \in [-1.66, +0.34]$: slightly shifted from mid-rapidity
- $y^* \in [-3.66, -1.66]$: significantly **forward** rapidities

This lets sPHENIX study **forward physics** without building a dedicated forward detector. Forward rapidity is where we expect to see interesting effects from the gluon-dominated regime (small-x physics) and saturation phenomena.
The shift $y_{\text{CM}} = 1.66$ is large enough to push the acceptance well away from mid-rapidity in the CM frame, which is exactly the goal of the asymmetric running.

Why $\eta = y$ is justified here

The problem explicitly states this assumption. The reasoning: for a particle with mass m, transverse momentum p_T, and rapidity y:

$$\eta - y = \frac{1}{2} \ln \frac{p_T^2 \cosh^2 y + m^2 + p_T \sinh y \sqrt{p_T^2 \cosh^2 y + m^2}}{p_T^2 \cosh^2 y + m^2 - p_T \sinh y \sqrt{p_T^2 \cosh^2 y + m^2}} \tag{33}$$

This difference vanishes when $m \to 0$. For pions ($m \approx 140$ MeV) at typical $p_T > 1$ GeV, the correction is small. The problem says to treat $\eta = y$ throughout.

Summary of All Results

Part	Result	Value
(a)	$y_{\text{CM}} = \frac{1}{2}\ln\dfrac{e^{y_a}+e^{y_b}}{e^{-y_a}+e^{-y_b}}$	(general formula)
(b)	$y_{\text{CM}} = \frac{1}{2}(y_a+y_b)$ for $m_a = m_b$	(simplification)
(c)	$y^*_{a,b} = \pm\frac{1}{2}(y_a - y_b)$ in CM frame	(equal & opposite)
5*(d)	$\sqrt{s} = 2\sqrt{E_a E_b}$	94.9 GeV
	y_a	$+6.28$
	y_b	-2.95
	y_{CM}	$+1.66$
	Left-end acceptance (CM)	$[-1.66, +0.34]$
	Right-end acceptance (CM)	$[-3.66, -1.66]$

The Big Picture

This problem teaches three things:

1. **Rapidity is additive under boosts** — this is its defining virtue and the reason collider physicists use it instead of angle or velocity.
2. **The CM rapidity is the average beam rapidity** (for equal-mass beams) — a beautifully simple result that follows directly from the additivity.
3. **Detector acceptance is geometric** but **physics is in the CM frame** — by choosing asymmetric beams and displacing the collision point, you can "slide" the CM-frame acceptance window to probe different rapidity regions without modifying the detector.